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Self-Adjusting Monitor for Measuring Aging Rate and Advancement

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ABSTRACT Time-variant age information of different parts of a system can be used for system-level performance improvement through high-level task scheduling, thus extending the life-time of the system. Progressive age information should provide the age state that the system is in, and the rate that it is being aged at. In this paper, we propose a structure that monitors certain paths of a circuit and detects its gradual age growth, and provides the aging rate and aging state of the circuit. The proposed monitors are placed on a selected set of nodes that represent a timing bottleneck of the system. These monitors sample expected data on these nodes, and compare them with the expected values. The timing of sampling changes as the circuit ages and its delay increases. The timing of sampling will provide a measure of aging advancement of a circuit. To assess the efficacy of the proposed method and compare it with other state-of-the-art aging monitors, we use them on selected nodes of the execution unit of different processors, as well as some circuits from ITC99 benchmarks. The results reveal that the precision of our proposed method is between 0.12 (ns) to 0.401 (ns). Its Area and power overhead are negligible and are about 2.13 and 0.69 percent respectively.

INDEX TERMS Age advancement, age monitoring clock, aging rate, self-adjusting monitors

I. INTRODUCTION

Technology scaling and transistor size reduction increase the uncertainty of various physical parameters like transistor threshold voltage. Although static variations are detectable after chip fabrication, dynamic variations could only emerge at the operational phase of the system. These variations result in circuits to operate at a slower speed than they were originally designed for. Two main physical phenomena that progressively lead to slowing down circuits are Bias Temperature Instability (BTI) and Hot-Carrier Injection (HCI), which are caused by stress on transistors due to the continuous movement of charges [1]–[3].

There are other phenomena that degrade the performance of a circuit. Electro-Migration (EM) increases the interconnect resistance, while Time-Dependent Dielectric Breakdown (TDDB) causes breakdown of transistor's gate [4]. These aging mechanisms have direct influence on the current and electrical parameters of transistors. Clearly, the aging process has important effects on the performance, complexity, and the overall cost of hardware.

Aging phenomena reduces the life-time and Mean Time To Failure (MTTF) of a circuit. Hence, the designer should take advantage of timing guard-bands (15 to 20 percent of

clock period [5]) for the design to alleviate these issues [6]. The timing guard-band mitigates the timing violations that are caused by increasing the delay of paths. Today's designers use a guard-band such that the circuits work for at least 7–10 years [2]. Thus, in this approach by estimating the impact of aging on delay of the circuit, a timing guard-band to satisfy the expected lifetime is considered. Hence, this approach leads to decreasing the performance of the system, especially in the early stages of circuit lifetime.

To reduce the guard-band and improve the performance, aging monitors that are placed in critical paths of the circuit are utilized. There is a type of aging monitor that allows the circuit to age, and when an error due to the aging mechanisms occurs, a method for recovery is followed. Voltage scaling and clock stretching are two common methods that are used after detecting the timing violation to recover the system [7]. On the other hand, there is another type of aging monitor that predicts the age of the system and prevents issuing any timing violation error. In this case, age of the system is predicted according to parameters that are monitored during the normal operation of the system. In this monitor type, delay degradations smaller than the considered guard-band are detected.

Age monitors that are placed in the circuit can be local or global. While local monitors are placed at critical paths in a circuit, global monitors are distributed throughout the circuit at random locations. A certain implementation of global monitors [8] uses specific structures (such as ring oscillator or replica circuits) and measures changes in frequency to estimate circuit aging. The philosophy behind the use of global monitors is low activity of selected nodes in local monitors that results delayed detection of system aging. Local sensors are used for a better accuracy and global sensors are added for a better reliability of age monitoring. Although our proposed method is a local monitoring system, adding a global monitoring system, improves its performance and is complementary to our method.

In this paper, we propose an *online, adjustable and configurable* monitoring structure that predicts the age of the system and detects *progressive changes* in delay, and provides the required information about the present state of the system and its aging rate. The work consists of a hardware structure for monitors, and analysis and simulations for placement of monitors in the circuit. The main features of this work in comparison with similar works done in this area are discussed below.

We perform *dual sampling* of the data propagated to monitored nodes. The first sampling is done with a clock that is generated by negative phase shifting of the system clock, and the second sampling is done by the main system clock. The two samples are then compared for consistency, and a flag is set otherwise. Sampling with a negative phase shift (before the active clock edge) is done to be able to predict the age of system before a failure occurs.

Dual sampling in this work is done by our innovative hardware structure for creating a *variable negative-shift phase clock*. This clock is configurable according to the age of system being tested. In addition, this clock is *self-adjusting* with parameters corresponding to process and environmental variations. This work can be compared with other negative-shift phase clock generation applications such as delay testing. Authors in [9] and [10] generate such a clock with FPGA PLLs. Low area overhead, adaptability, and variability of our hardware are some of the unique features of our clock generation. This proposed structure has been modeled in SPICE and its functionality has been verified.

Another challenge in today's technology is dealing with *voltage and frequency multiplicity*. None of the works on age correction or age monitoring in the literature handle this issue. However, we address *voltage and frequency multiplicity* by a value in our negative-shift phase clock generation hardware that is adjusted based on supply voltage and frequency of operation. A register file in our proposed mechanism saves age counters for various voltage-frequency value pairs.

An important aspect of our work that distinguishes our age sensing mechanism from other works is providing *the rate of aging* as well as the age of the system. The rate at which a system is aging is a very useful parameter for allocation of tasks and balancing activities in a multi-processing system. The age sensing hardware proposed in this work saves age

related information in a group of counters (registers), based on which age of system and rate of aging can be calculated. Usually, this information is provided to an upper-level task scheduler and calculation is performed at that level.

Last, but not least, important factor in our work is selection of internal circuit nodes for placement of aging monitors. Such monitors are placed on paths in the circuit that are more likely to age. The number of monitors is optimized considering the expected life-time of the circuit and an allowed area/power overhead. An optimization method is utilized that considers various age and performance parameters for finding best nodes for monitor placement.

In this work, elaborated simulations at the circuit (SPICE) and gate levels (Verilog) have been performed for verifying sensor functionality, age detection verification and calculation of hardware/power overhead considering monitor precision. In addition, several benchmark circuits have been synthesized, and analyzed for the delay and allowed guard-band values. These circuits are used in the evaluation of our age detection mechanism.

In the rest of this paper, Section II presents related works and discusses several monitoring methods in the literature. Section III presents details of the proposed age monitoring structure. Critical paths and monitoring nodes selection and the structure of internal node monitors are discussed in Section IV. The experimental results are presented in Section V. Section VI is devoted to a discussion of how various parameters such as noise, process variation, supply voltage changes and fluctuations, and temperature affect our delay measurements and age predictions. This is followed by the conclusions in Section VII.

II. RELATED WORKS

In this section, some methods in literature for age detecting and predicting will be discussed. Most of the studies focus on monitoring delay degradation due to aging phenomena. However, there are other works that try to measure certain physical parameters. Measuring the reduction of transistor's saturation current, and a structure that measures the voltage for turning-on a transistor are among such works [11]. These works usually include analog parts and cause difficulty in design and synthesis of digital circuits. Also, test of these components is a hard task and increases the test time and manufacturing costs. These methods are categorized as the age detection methods that only detect failure of the system. Some of the better known works of this category are Razor [12], [13] and the age sensor in [14]. Methods in [13] and [14] can only detect the delay failure using a delayed capture of the output value, and then they correct them. However, these methods have a low hardware and power overhead and are suitable for the applications that have area and power overhead concerns.

Other works in the area of aging are based on replicating certain critical regions of the circuit and comparing the delay of identical paths (from the original and the replicated circuit) to detect the aging effects [7]. Inequality of delays between these identical regions indicates system aging. Replica [7] is

one such method. The problem with these methods is the possibility of having a mismatch due to the process or environmental variations, between Replica circuit and the original one, which results in wrong age monitoring. To reduce the mismatch, Tunable Replica has been proposed [15]. In Tunable Replica circuits, all types of variations have not been considered and also, these circuits cannot detect the gradual changes in delay of the circuit.

There are other methods that insert a circuit into the system that mimics delay behavior of critical regions. Some examples are [16] and [17] that use Ring Oscillator (RO) to emulate the delay behavior of the critical regions. These methods have the same problems as those of the replication methods. Area overhead of RO based age detection circuits is less than the Replica methods. In addition, measuring delays of RO based circuits is simpler and more accurate than the latter.

It is also possible to use some parts of the hardware that are already embedded in the system for monitoring purposes. For example, we can use the boundary scan cells for monitoring the path delay of a circuit in its normal operation mode. Hence, in [18], it has been suggested to use the structure for the Small Delay Defect (SDD) testing (that is already in the system) for predicting the system's age. This method is the first work that utilizes the age detection circuit for other purposes. This method is interesting and tries to predict circuit's age in the operational mode. The method creates a time interval that checks any transition on output nodes in a specified time window. The proposed structure utilizes delay elements for generating a second clock. The width of the specified time interval cannot be changed during normal operation of the system. Therefore, this method cannot predict the progress of the system's age. Also, the method cannot adjust itself to the progressive age of the system. Moreover, for any age detection, they use one of the generated aging clocks instead of the main clock in order to capture main data. This can corrupt the main data and causes failures in the system. Also, inserting extra hardware on the main clock path can generate skew, jitter, and noise on the main clock, thus creating new errors in the system. In addition, the method does not examine different voltage and frequency states of the system. The effect of process variation on the delay of critical paths and time interval is not considered in this work.

In [11], a sensor for detecting system's age is proposed. The authors propose a built-in self-sensor that estimates the time between changes at critical output nodes. The age of the system is determined according to this time. The method detects the age of a system when it exceeds the period of the main clock and therefore cannot predict the aging growth. In addition to these issues, the sensor does not work at the system normal operation, and therefore it can age the circuit at the offline time.

The online aging prediction has attracted much attention in recent years. Agarwal *et al.* have proposed the design of an aging sensor in [6], wherein a warning signal will be triggered if a delayed transition occurs within the predetermined time interval. This aging sensor is one of the most popular circuits that are used in the works for reducing the aging effects.

In [19], a similar method to that of [6] is proposed. It uses a stability checker that monitors any transition in a unified time interval before the rising edge of the clock. This work has some problems that are consistent with previous works that only detect transitions in a time interval, without pointing to the exact time of occurrence. This method has a precise age detection.

In [20], authors claim that their method can predict the aging rate of the system at run time. The method deactivates the system at a different time interval and takes a snapshot of the system on scan flip-flops and constructs a signature. A software predicts the age of system according to the signatures and thermal situation. In this method, the state of the system after chip fabrication that is different due process variations is not considered. This method interferes with the normal operation of the system; therefore, it will degrade the performance of the system. Utilizing a learning software can decrease the area and power overhead of the aging prediction circuits. Because this method does not add any monitoring hardware to the circuit's path, it does not increase the clock period of the system.

In [8], [21], a monitoring method is introduced that considers both local and global monitoring. This method improves the accuracy of age monitoring. The main difference of our work with [8] is that our method detects system's age before the active edge of the clock and it also measures the aging rate. We do not address global monitoring [8], and it can be adopted and adapted to our techniques to complement our elaborate local age monitoring.

Most of the age monitoring methods monitor the end point of critical paths, resulting in the loss of accuracy in delay measurement. Monitoring the internal nodes of the paths instead the output one is first introduced in [22]. The slack probe methodology in [22] inserts monitors in the internal nodes of the circuit like probes. This method helps to detect delay failures of timing variation defects as soon as they occur. Monitoring the internal nodes can show changes in delay faster than what is possible by monitoring output ports. Their monitoring technique is constructing a chain of gates after selected internal node and checking the delay of this chain. This technique has a high area overhead and a difficult synthesis phase.

The above discussions highlight some of the main features of existing age sensing mechanisms. Several of these works are evaluated in Table 1 for the features they support. In this paper, we have implemented age prediction using adjustable sensors that use negative-shift phase clock that is self-adjustable in band-gap detection, and adaptable to environmental variations. The mechanism of monitors we use and placement of the sensors in the circuit provide features that have not been addressed in any of the previous works. Our proposed mechanism is an online (Column 4 in Table 1) method that predicts aging before it actually occurs. Our sensor structures are initialized to account for process variations (Column 7 in Table 1) and adapt themselves to future variations due to aging. An important feature not found in any of the previous works is

TABLE 1. The comparison of our method capabilities with the previous works.

Features Methods	*Detect or Predict	Predicting aging rate	Online	Interfering with normal operation	Can be used for SDD test	Considering process variations	Handling multi-voltage domain
Our method	P	✓	✓	×	✓	✓	✓
[6]	P	×	✓	×	×	✓	×
[18]	P	×	✓	×	✓	×	×
[11]	D	×	×	✓	✓	×	×
[12], [13]	D	×	✓	×	✓	×	×
[19]	P	×	×	✓	×	×	×
[20]	P	✓	×	✓	×	×	×
[7], [15], [16]	D	×	✓	×	×	×	×
[22]	D	×	✓	×	✓	×	×

*P: Prediction, D: Detection

handling multi-voltage domain of system components (Column 8). The monitors are at the gate-level, which makes them easy to use with standard cell designs, and does not require detailed transistor-level considerations. All of these make our work unique in this kind. The proposed idea can also easily be used for the SDD testing. By applying delay test patterns in the offline mode and activating the monitor structure, the SDD test is performed and defects will be detected.

III. AGE MONITORING STRUCTURE AND OPERATION

The heart of our age monitoring system is the monitoring structure that we use. This structure implements features shown in Table 1 that distinguish our work from other works on aging. The structure and how it is made to implement columns shown in the table are discussed in this section.

Different paths of a system have different delays and therefore different slack times with respect to the system clock period. The slack time is defined as the amount of time that the output value of a path can be delayed without violating the active edge of the clock, and the correct value is captured in the corresponding flip-flop. Therefore, delay of the path, setup time, Clock to Q time and the defined guard-band are considered for obtaining the slack time of a path. Aging causes reduction in this slack time. A system begins with a starting slack time larger than zero. As the circuit is aged, its slack time is reduced, causing it to fail when the slack time reaches zero [1], [2]. A digital component ages faster in its early life stages, therefore for a more intelligent prediction method, the starting slack time at birth and variation in slack times during the operation of the circuit are needed in order to prevent system failure.

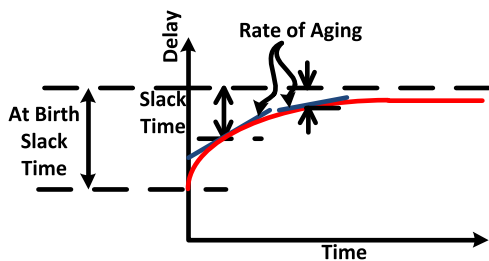


FIGURE 1. Logic delay changes during circuit's lifetime.

Figure 1 shows how an initial slack time due to process variation, slack time at a given point in time, and the rate at which slack times change are related. To be able to predict this exponential function and make proper age-related decisions, the initial slack time and time-dependent age of a system are required. Using this information, the rate of aging can be calculated. In our proposed hardware structure, counters are provided in the monitor circuit that are initialized with values that represent the initial timing of the circuit after it has been manufactured. For every operating voltage, these counters keep track of the slack time at various time instances.

The overall structure of the proposed monitoring method is shown in Figure 2. The dotted areas in this figure show the combinational part of the system between registers i and j (R_i , R_j). Registers use the system clock, and we assume a 50 percent duty cycle clock with rising-edge trigger registers. To measure circuit's age, the critical paths of the combinational part should be determined which are indicated by the critical path cluster in this figure. A cluster is a small subset of a combinational cloud of a circuit that is sandwiched between two layers of registers. A critical cluster contains critical paths with nearly equal slack times.

The proposed age monitoring uses a clock that lags behind the system clock by the slack time of the critical path cluster. This clock, that is called AM-Clock, is used for early capture of

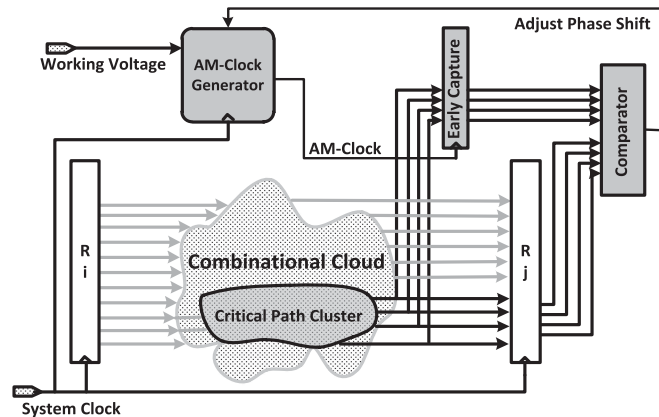


FIGURE 2. Overall monitoring structure.

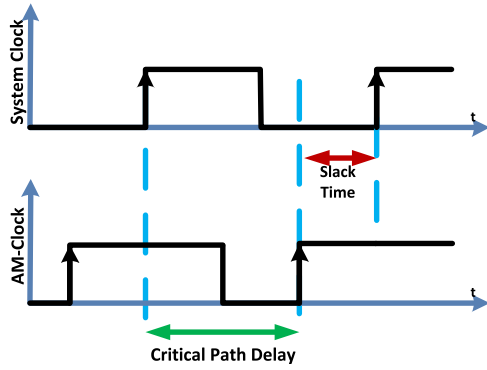


FIGURE 3. Phase difference between the system clock and the AM-Clock for a critical path cluster.

the data from the critical path cluster. Figure 3 shows the timing diagram of the system clock and AM-Clock. The rising edge of AM-Clock follows the slack time of the critical path cluster. AM-Clock is generated by a circuit that we refer to as the AM-Clock generator that will be explained later in this section.

Initially, the at-birth delay of the critical path cluster is measured, and AM-Clock is set accordingly. Every time that the age monitoring is activated, the output of the critical path cluster is sampled by two clock edges; the system clock (in Rj register in Figure 2) and AM-Clock (in Early Capture Register (ECR) in Figure 2). Then, the contents of the Rj and ECR are compared. The register contents being equal indicates that the slack time of the critical path cluster has not changed. However, when the contents of the Rj and ECR are different, a comparator sends the Adjust Phase Shift (APS) signal to the AM-Clock generator to adjust AM-Clock. In the next several clock cycles, this process is repeated until the APS signal is no longer issued, in which case adjusting has been completed. Based on phase difference of the system clock and AM-Clock, the aging rate and the current slack time are estimated.

The gray areas of Figure 2 illustrate parts that belong to our proposed monitoring structure and should be added to the circuit for each critical path cluster. The detailed structure of these parts and their functionalities are provided in the following sub-sections.

A. EARLY CAPTURE REGISTER (ECR)

Early capture register is a simple register that is clocked with AM-Clock and captures data that arrive at the monitoring nodes before the rising edge of the system clock. This is the first sampling mentioned above.

B. SAMPLE COMPARATOR

The sample comparator unit is a simple circuit (XOR gates) that compares its two inputs from ECR and Rj, and issues the APS signal in the case of inequality. While ECR has the first sampling of the monitoring node values, Rj outputs have the second sampling done by the system clock edge.

C. AM-CLOCK GENERATOR

The main part of the monitoring structure is the AM-Clock generator. The internal structure of the AM-Clock generator

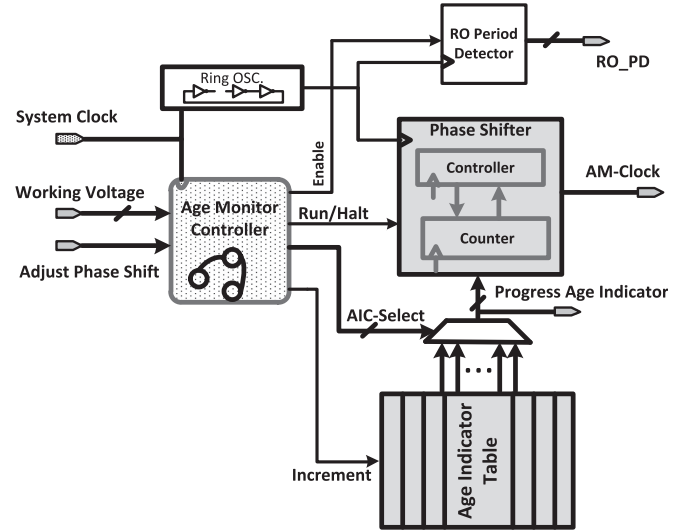


FIGURE 4. Structure of the AM-Clock generator.

is shown in Figure 4. AM-Clock generator contains five components: Age Monitor Controller, Ring Oscillator (RO), Phase Shifter, Age Indicator Table (AIT) and RO Period Detector (will be discussed in Section III-G). The inputs of this unit are system clock, APS signal, and the operating voltage index. The operating voltage index is a binary data that shows the index of the current operating voltage of the system that is determined by the Dynamic Voltage Frequency Scaling (DVFS) unit (not part of this work). Also, the outputs of the AM-Clock generator unit are AM-Clock, Progressive Age Indicator (PAI), and RO_PD (ring oscillator period detector) index. PAI shows the instantaneous age of the system. The monitoring structure provides a counter for every operating voltage of the system. The counters are referred to as Age Indicator Counters (AI-Counter, or AIC) and are stored in an Age Indicator Table (AIT). For a given operating voltage level, the content of the corresponding counter is read, and it provides the progressive age information (PAI) at that voltage level.

In this structure, RO generates a clock that is synchronized with the system clock, but with a higher frequency. The phase shifter unit uses the clock generated by RO to shift the phase of the system clock in order to generate AM-Clock. The amount of phase shifting is determined by AIT. Different parts of the AM-Clock generator will be discussed in the following sections.

1) AGE MONITOR CONTROLLER

The age monitor controller has two inputs and three outputs. The controller waits for an APS signal from the comparator of Figure 2. When received, it halts the phase shifter, and issues an *increment* signal to increase all counters of AIT by one. Then, it resumes the phase-shifter operation. Also, based on the voltage-frequency index, the controller selects one of the AI-Counters output to be used by the phase shifter unit. In addition, when the APS signal is issued, the *Enable* signal of the RO period detector becomes high.

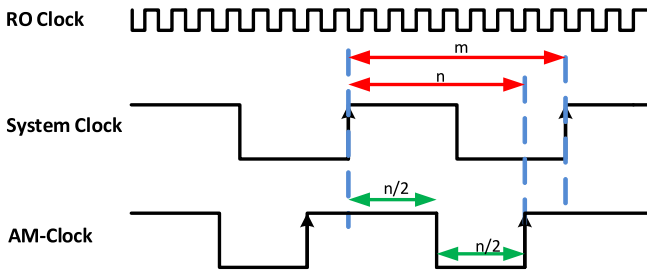


FIGURE 5. Phase shifter waveform.

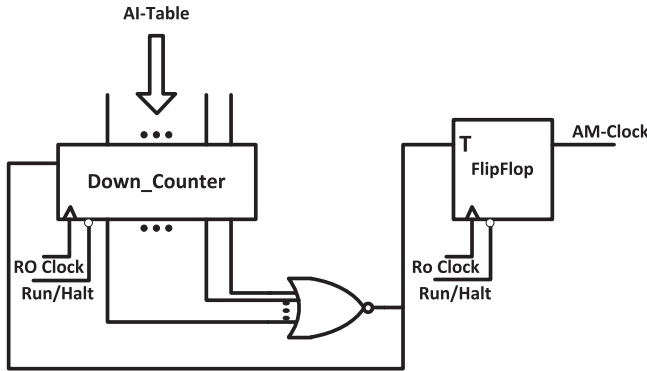


FIGURE 6. Phase shifter structure.

2) PHASE SHIFTER

The phase shifter unit takes the system clock as an input and generates AM-Clock by shifting and changing the duty cycle of the system clock. The amount of shifting is described in terms of number of RO clock cycles. Figure 5 shows the system clock and AM-Clock that is shifted n RO cycles. This figure also shows that the period of the system clock is almost equal to m RO cycles.

The phase shifter has two operating modes, *halt* and *run*. In the *halt* mode, AM-Clock is frozen which happens when the amount of shifting should be changed or the monitoring unit is deactivated. On the other hand, when the phase-shifter is in the *run* operating mode, it waits for the rising edge of the system clock and produces the falling edge of the AM-Clock $n/2$ RO cycles later. The next rising edge of AM-Clock is generated $n/2$ RO cycles after the falling edge. Hence, the AM-Clock rising edge is the shifted rising edge of the system clock, while its duty cycle is changed to $(m - 0.5n)/m$.

The internal structure of the phase shifter is shown in Figure 6. Phase shifter gets the AIT output, run/halt signal (from the controller) and an RO clock (a high-frequency clock) as its inputs. The circuit consists of a down binary counter and a T flip-flop. In this circuit, when the run/halt signal is 0 (i.e., halt), the counter and T flip-flop will be reset. The flip-flop content at the reset is 1. When the run/halt is 1 (i.e., run), down-counter and flip-flop are at the operational mode. At each rising edge of the RO clock, the counter counts down, and when it reaches 0, the input of T flip-flop will be 1 and its stored value will be inverted. In this state, the falling edge of AM-Clock will be generated. Then, the counter will be loaded with the initial value (The AIT output is considered as the initial value of

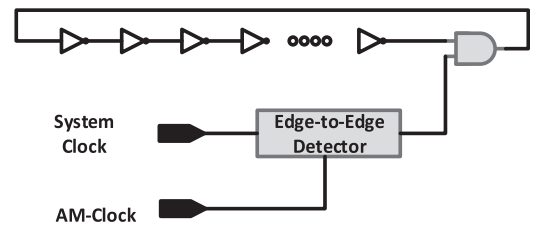


FIGURE 7. RO and synchronizing with the system clock.

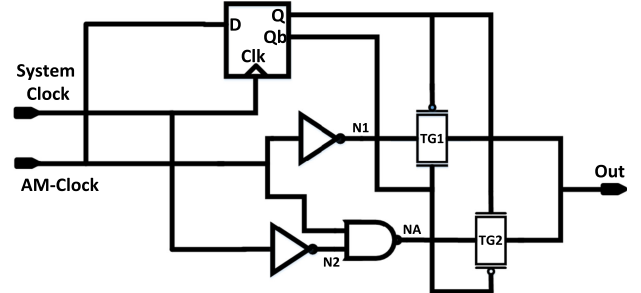


FIGURE 8. Structure of the Edge-to-Edge detector.

TABLE 2. Operation states of the edge-to-edge detector.

Am-Clock	System Clock	N1	N2	NA	Q	TG1	TG2	Out
0	0	1	1	1	1	off	on	1
0	↑	1	0	1	1	on	off	1
0	↓	1	1	1	0	on	off	1
↑	0	0	1	0	1	off	on	0
1	0	0	1	0	1	off	on	0
1	↑	0	0	1	1	off	on	1
1	1	0	0	1	1	off	on	1

the down binary counter), and the above operations are repeated. When the counting reaches 0, the state of T flip-flop is changed and the rising edge of AM-Clock is generated.

3) RING OSCILLATOR

The RO unit generates a high-frequency clock signal that is used as a reference for shifting the system clock. The internal structure of this unit is depicted in Figure 7. This structure generates a synchronized RO clock with the system clock. The proposed RO block oscillates only between the rising edge of the system clock and the rising edge of AM-Clock. The Edge-to-Edge detector unit manages synchronization and oscillation. RO oscillates when the output of the Edge-to-Edge detector is high. Figure 8 shows the internal structure of this block. The operation states of this circuit are summarized in Table 2. SPICE simulation waveforms of this circuit in 45 nm NanGate technology [23] are shown in Figure 9. The solid line waveform in this figure is the output of the Edge-to-Edge detector. As shown, the output of this circuit is high between the rising edge of the system clock and the rising edge of AM-Clock.

4) AGE INDICATOR TABLE (AIT)

The AI Table output determines the amount of shifting of the rising edge of the system clock to generate AM-Clock. This

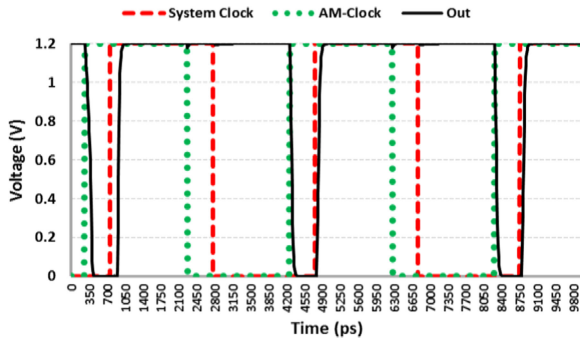


FIGURE 9. SPICE simulation waveform of the Edge-to-Edge detector.

unit is for adjusting and reconfiguring the monitoring structure according to the aging progress, environmental parameters, and process variations. AIT consists of several AI-Counters, each of which corresponds to one of the working voltage levels of the system. The number of the levels is determined by the DVFS unit of the system. As an example, if the DVFS unit only considers four operating voltage levels, AIT includes four AI-Counters all of which are incremented simultaneously when an APS signal is issued.

The width of AI-Counters is determined such that counters can hold values calculated by the ratio of the system clock period over the RO clock period.

D. MONITOR INITIALIZATION AND OPERATION

This part discusses the method that determines the at-birth value of the AI-Counters. We also discuss the online operation of the monitoring structure after it has been initialized.

The delay of the longest path in the selected critical cluster will determine the amount of AM-Clock shifting with respect to the system clock (parameter n in Figure 5 that we refer to as the at-birth value). A path (from critical paths set) starts from one of the inputs and terminates with one of the selected nodes for monitoring. The delay of the path is estimated at the design time with timing analysis tools such as Synopsys Design Compiler. The at-birth values are calculated by the ratio of the amount of shifting AM-Clock over the RO clock period. Hence, the at-birth values of AI-Counters strongly depend on the RO frequency. An RO that generates a faster clock results in a more precise delay measurement and better detection of system aging. Therefore, the maximum operating frequency of AI-Counters must be considered as the frequency of the RO clock.

In this work, during the design time, we estimate the at-birth values of the critical clusters under different voltage levels and store them in the system memory. These values are loaded into the AI-Counters at system boot time through its scan chains.

In the runtime, at each operating voltage level, the value of its AI-Counter must be increased when the APS signal is asserted. However, in this case, the proposed monitoring system increases the values of other AI-Counters. We suggest to perform this because when a path is aged in an operating

voltage level, its slack times in other operating voltage levels are also reduced. Also, based on our studies when the value of an AI-Counter is increased, the values of other AI-Counters should at least be incremented by one. It is important to mention that when the AI-Counter value shows a smaller value than the actual delay of paths, the APS signal becomes one, and the value of the AI-Counter will be adjusted accordingly.

During the runtime of the system, we suggest to periodically store the current values of AI-Counters in the system memory that contains information about the monitoring system. Therefore, the contents of the counters are stored before power off. Mechanism for restoring other information from memory (during the system boot) will be used for counter value re-fetching or recovering.

E. IMPACT OF TEMPERATURE AND VOLTAGE ON THE MONITORING STRUCTURE

The above section explained our approach for extracting the at-birth values of AI-Counters during the design time. Since temperature variations and the operating voltage impact the slack time of circuit logic paths, their effects should be considered in finding proper values for the AI-Counter. An increase in temperature affects circuit delay and the AM-Clock generator. Since AI-Count values are based on ratios of circuit delay and RO frequency, our experiments show that counter values remain unchanged with changes in temperature.

The impact of the temperature by altering the operating voltage level on the monitoring functionality should be examined. To show the impact of the temperature variation under different operating voltage levels, we have simulated the execution unit of the 16-bit ARM processor and the RO unit of the proposed monitoring system by SPICE simulation in 45 nm technology under different voltage levels and temperatures. In this example, we assumed that the DVFS unit selects the operating voltage from a set of voltages with 9 operating voltage levels (i.e., {0.8 V, 0.85 V, 0.9 V, 0.95 V, 1 V, 1.05 V, 1.1 V, 1.15 V, 1.2 V}). Because of the small size of the monitoring circuit and also its placement near the critical paths, we consider the same temperature for circuit logic paths and monitoring circuit. We also assume that change in voltage imposed by DVFS is applied to the entire circuit (including the monitors). This is a correct assumption, because the sensors are small parts of the circuit and their voltages are not handled separately. Figure 10(a) and 10(b) show delay variations of the execution unit of the ARM processor and the RO unit, respectively. As shown in these figures, delay values of paths decrease in higher voltages and increase for higher temperatures.

We also extract the at-birth values of 9 operating voltage levels under different temperature conditions. As Figure 10(c) shows, the at-birth values are independent of temperature values for each operating voltage level. This originates from the fact that changing the period of the RO clock is relatively proportional to delay change of the critical paths under temperature variations. On the other hand, Figure 10(c) shows changing in the operating voltage changes the at-birth value. We handle this variation with AI-Counters in AIT.

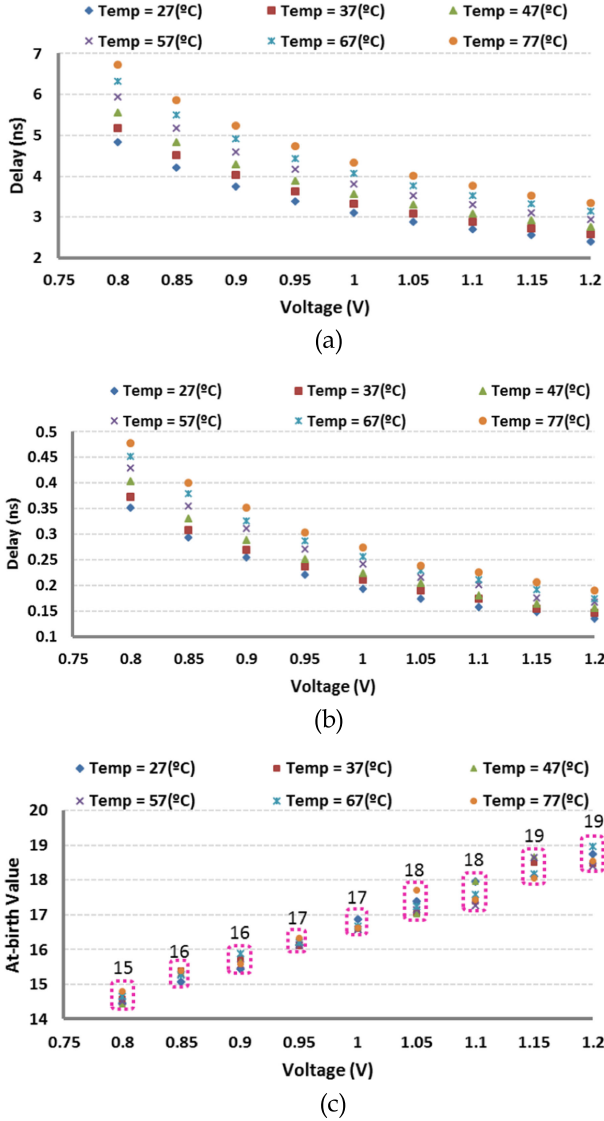


FIGURE 10. (a) Delay versus operating voltage level at different temperatures for critical path of the execution unit of ARM, (b) Delay versus operating voltage level at different temperatures for RO unit, (c) At-birth value for different operating voltage levels and temperatures.

F. IMPACT OF PROCESS VARIATION ON THE MONITORING STRUCTURE

As mentioned before, during the design time, the at-birth value of the AI-Counters are determined. However, because of process variation effects on the path delays, the extracted values at design times are not reliable. If the process variation increases the delay of a path, our monitoring system is able to automatically adjust the at-birth value (the estimated at-birth value is smaller than the actual at-birth value). On the other hand, when the path delays become shorter due to the process variation the estimated at-birth value is larger than the correct at-birth value. Therefore, the proposed monitoring system is not able to find the correct at-birth value. Hence, for considering process variation, we scale down the estimated at-birth values that are determined at

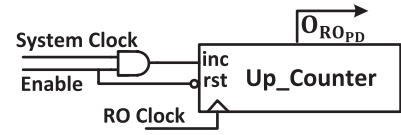


FIGURE 11. Structure of the RO period detector.

the design time by multiplying them by a parameter α (where $\alpha < 1$). This parameter is to be calculated at the design time by estimating the impact of the process variation. We estimate the α parameter based on the worst effect of process variation on the critical paths delay. This guarantees correct age measurement for all critical paths with different amounts of process variation. As an example, if we know that the process variation in the worst-case decreases the delay by a factor of $0.8\times$, the parameter is set at 0.8.

After circuit fabrication and during the testing phase, the scaled at-birth values are stored in AIT. Next, by applying different operating voltage levels and activating the monitoring structure, it automatically adjusts the values of AI-Counters. The impact of process variation on the monitoring structure and its precision will be discussed in Section VI.

G. ESTIMATING SLACK TIME AND AGING RATE

By employing the current values of the AI-Counters and the rate of changes, the present state of the slack time and the rate of changing in the slack time (aging rate) can be estimated. The present slack time is calculated using the clock period of the system clock ($CP_{SystemClock}$), the clock period of the RO (CP_{RO}), and the current value of AI-Counter. The clock period of the RO unit is changed due to the process variation and aging. Hence, for indicating the clock period of the RO unit, the RO period detector unit is used. The internal structure of this unit is depicted in Figure 11. This unit includes a counter that counts when the system clock and the *Enable* input are high. In this case, based on the output of this counter (O_{ROPD}) and by assuming that the duty cycle of the system clock is 50 percent, the CP_{RO} is equal to $\left(\frac{0.5 \times CP_{SystemClock}}{O_{ROPD}}\right)$.

Now, the current slack of the path is determined by (1).

$$Slack_{Current} = CP_{SystemClock} - (CP_{RO} \times O_{AI}) \quad (1)$$

In (1), O_{AI} is the value of AI-Counter of the corresponding operating voltage level.

By tracking the changing of the slack of paths, their aging rate can be estimated. In early lifetime of the circuit, the aging rate is higher; hence, the interval between the tracking processes should be tighter. When the monitoring circuits are not used, the system clock can be gated to reduce power consumption of the monitoring structure. Also, in the cases where the interval is large enough, the monitoring system may be powered-off by power gating.

This section showed how monitors are constructed along with their hardware, storage for saving various parameters, and how reported information of monitor structure is utilized. The next section shows where these monitors are to be placed.

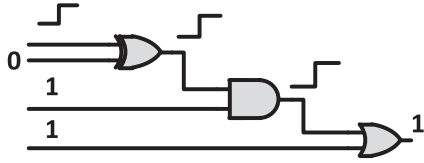


FIGURE 12. Masking of the transition in the internal nodes.

IV. AGING CRITICAL PATHS AND MONITORING NODE SELECTION

The best place to insert sensors for age detection is along circuit critical paths, or on paths that are closely related to the critical paths. This section discusses our methodology for selection of circuit nodes for age monitor placement.

Although the longest paths in a circuit and their corresponding delay values determine the circuit clock period, variations and aging do not necessarily happen on circuit longest paths. Therefore, some circuit paths can become critical due to static and dynamic variations. We use our method in [24] and [25] for critical path selection, simultaneously. In [24], longest paths are selected first as the Potential Critical Paths (PCPs). Then according to random variation analysis, some of other paths are also considered as PCPs. In [25], we select critical paths considering BTI, HCI, and temperature.

Monitor structure should find changes in delay of the critical path as soon as possible. Because of low activity (not always) on the critical paths, monitoring critical path output activities does not necessarily give us enough incidents to measure the aging effect. Because of this, instead of selecting end point of critical paths for age monitoring, we select mid-points in the circuit leading to critical paths. This way, we monitor more activities on the internal nodes than having to wait for input transitions to reach the output nodes, fewer of which will survive. This is because as gates converge towards circuit outputs and as more gates are involved, transitions are masked, and some never reach circuit outputs. Therefore, there will be a lag in the detection of the change of delay on circuit outputs. An example of masking signal transitions in internal nodes is shown in Figure 12. As illustrated in this circuit, three transitions occur through internal nodes, but they are masked with the OR gate.

For selecting internal nodes of the critical path, we use our work in [26] that selects nodes according to properties that we discuss here. We selected the most shared and active nodes and also consider other properties such as electromigration in input cone, the sensitivity of the input logic cone, etc. in PCPs for monitoring. The most shared nodes are selected to minimize the number of monitoring nodes, thus minimizing the hardware and power overhead of the monitoring system. Since tracking aging is best done on nodes with high activity, we have used node activity as a parameter for selecting nodes for monitor insertion.

We also consider the performance loss due to monitoring hardware in our node selection methods and try not to select nodes of longest paths, because such nodes impose extra loads on the longest paths adding to critical delays. Instead

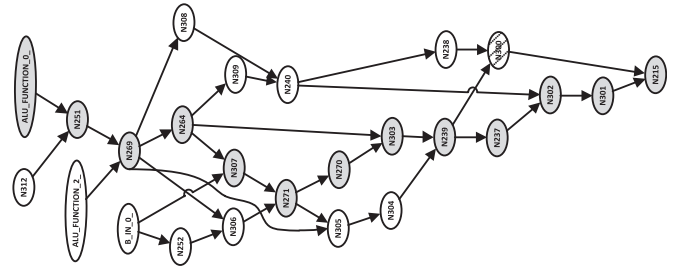


FIGURE 13. Node selection for monitoring in non-longest paths.

we look for shorter paths that have a relatively high overlap with the longest paths, and place our selected nodes on the non-overlap part (end tail) of the shorter paths. The shorter path covers most of the nodes of the longer path, but leaves some nodes not monitored for the price of higher performance (if required). If degradation in performance can be tolerated, finding overlapping shorter paths can be skipped.

For example, consider the circuit in Figure 13. This figure shows a critical region of the 4-bit PLASMA processor. In this circuit, we select node N300 (patterned line node) that is not in the longest path (gray nodes). With a good approximation, path leading to this node represents the longest path in the graph. The longest path has 13 nodes, and nine of them are covered by inserting a monitor at node N300. If we consider the same situation of node N237 with N300, then 76.9 percent of the longest path nodes will be covered without any performance loss on the longest path. It should be noted that it is not possible to find this situation in all regions or all circuits.

After node selection for monitoring, we should add the monitoring structure to the selected nodes. For this purpose, the structure (gray parts) shown in Figure 14 are inserted at selected nodes. In this structure, a flip-flop that is clocked with AM-Clock is inserted at the selected node (e.g., the circled node). The flip-flop captures value of the node at the rising edge of this clock. The value of the node is also sampled after the AM-Clock edge and compared with the captured value in the flip-flop. If a mismatch occurs in any selected node, the APS signal will be activated.

The number of monitored nodes that can summarize in one OR gate and one flip-flop is limited. Therefore, Equation (2) should be satisfied, and according to the delay of the OR gate, we can obtain the number of monitored nodes for each flip-flop.

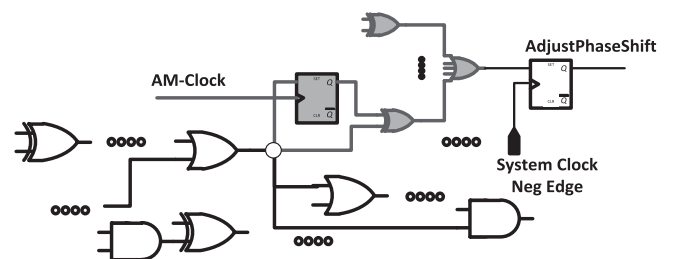


FIGURE 14. Monitoring structure at the selected internal nodes.

TABLE 3. Features of the examined benchmarks.

Circuits	#Inputs	#Outputs	#Gates	#FF	#Paths	#MN	$F_{SystemClock}$	F_{RO}	#inv
P2	59	41	13850	95	>22000	18	0.29	4.67	5
P3	231	71	5183	320	>31000	10	0.36	4.67	5
P4	258	235	32500	491	>100000	40	0.101	2.63	9
P5	149	190	35612	533	>100000	35	0.109	2.63	9
C1	5	6	963	121	>20245	5	0.48	4.67	5
C2	36	70	7922	449	>100000	7	0.18	4.67	5
C3	32	54	8567	245	>100000	13	0.21	4.67	5
C4	37	97	27825	1415	>100000	17	0.18	4.67	5
C5	37	23	94249	3320	>100000	24	0.14	2.63	9

$$t_{xor} + t_{or} < t_{SD} < \frac{t_{clk}}{2} \quad (2)$$

In this equation, t_{xor} is the delay of a 2-input XOR gate, t_{or} is the delay of an n -input OR gate, in which n is an unknown variable, t_{SD} is delay of the shortest path in input cone of the monitored node, and t_{clk} is the system clock period. The delay of the shortest path is important to prevent the data overrun from the next clock cycle. Solving this equation, the n will be obtained.

V. EXPERIMENTAL RESULTS

To assess the efficacy of the proposed monitoring circuit, we applied it on the critical paths of the execution stage of several pipelined processors and also five circuits from ITC99 benchmarks. The set of the selected benchmarks includes 4-bit Mini RISC, 16-bit MIPS, 16-bit ARM, 32-bit Alpha processor, and 32-bit MIPS(P1-P5) and, b12, b15, b14, b17 and b18 circuits (C1-C5) from ITC99 benchmarks. Features of the examined benchmarks are summarized in Table 3. In this table, #Paths is the number of circuit paths, #FF is the number of flip-flops in the circuits, #Gates is the number of circuit gates, #MN is the number of selected monitoring nodes, $F_{SystemClock}$ is the frequency of the system clock (GHz), F_{RO} is the frequency of RO clock (GHz), and #inv is the number of inverter elements in the RO unit. For this study, we first synthesized the HDL model of the benchmark circuits, and next by employing an in-house tool, their SPICE models were extracted. Then, Synopsys HSPICE tool in 45 nm technology was used to investigate the effectiveness of the proposed monitoring system. For aging impact modeling, we have used MOSRA (MOS Reliability Analysis) feature of the HSPICE tool. Note that the SPICE model of the monitoring structure was also used for verifying the functionality of the proposed structure and obtaining power consumption. Based on MOSRA reports, under the applied workload, the delay of critical paths of the considered benchmarks increases 10 to 17 percent after ten years.

A. LOW LEVEL SIMULATION OF MONITORING STRUCTURE

For verifying the monitoring structure, we initialized its AI-Counter with a value smaller than the estimated at-birth value at the design time. Then, we simulated the SPICE model of the circuit with the inclusion of the monitoring structure. Initially, AM-Clock was smaller than the expected value and it

was close to the situation of aging delay on the critical paths. In all simulations, the APS signal was issued and AI-Counters were adjusted to their new values. This observation was continued until AI-Counters reached their at-birth values. This procedure was performed for nine mentioned working voltages, and the results verified the correct functionality of the monitoring structure.

B. EVALUATING EFFECTIVENESS OF MONITORING NODES

In order to evaluate the effectiveness of our approach for selecting monitoring nodes, we performed delay fault simulation on the studied benchmark circuits. This study mimicked the aging-induced delay degradation of the critical paths by injecting delay faults. For this purpose, delay degradation of the critical paths has been extracted by MOSRA feature of the HSPICE tool.

We injected delay faults (using an in-house Verilog PLI (Programmable Language Interface) function tool [27]) in random places of the PCs. About 5K random data patterns were applied to the input of benchmark circuits. Simulation results showed that, on average, about 96.16 percent of the injected delay faults have been detected at selected nodes for monitoring. However, about 2 percent of the injected faults were not detected at the selected nodes. Our selection of nodes is further verified by the fact that these delay faults were not detected at any of the nodes in the circuit with the same distance as our selected nodes to the inputs of the circuit, or nodes that were closer to the outputs. These undetectable delay faults were injected at the level near the inputs and were masked with some gates of the paths.

About 1.84 percent of the injected faults were injected at the level higher than the level of the selected nodes, and hence, they were not detected. About 0.8 percent of these faults were masked before reaching the output nodes of the circuit, while the other faults (about 1.04 percent) were detected at the output nodes. Note that undetected delay faults do not pose a major problem, because in this study single delay faults were injected, while in an actual situation, the delay degradation of a critical path is the summation of the increased delay of all its gates.

To perform a comparison study, we performed the above fault simulation procedure for the circuits if only the outputs of the critical paths were selected for monitoring. We applied same 5K random data. In this case, on average, only 63.7 percent of the injected delay faults were detected at the output nodes. The other injected delay faults, about 36.3 percent, were masked through the gates of the paths. Therefore, the simulation results show the efficacy of the proposed monitoring internal node selection approach.

C. AGE DETECTION PRECISION

Age detection precision of our proposed monitoring method depends on the width of the AI-Counters and the period of the RO clock. The minimum width of AI-Counters is defined based on the slack time of critical paths by considering the

TABLE 4. Counter width and precision (ns) of monitoring structure based on guard-band (%) and working voltage (V).

Circuits	Guard-band	Voltage/ AIC width	0.8	0.85	0.9	0.95	1.00	1.05	1.1	1.15	1.2
C1	0<GB<20	5	0.214	0.204	0.211	0.188	0.174	0.152	0.137	0.125	0.112
C2	0<GB<20	5	0.214	0.204	0.211	0.188	0.174	0.152	0.137	0.125	0.112
C3	0<GB<20	5	0.214	0.204	0.211	0.188	0.174	0.152	0.137	0.125	0.112
C4	0<GB<=15	5	0.214	0.204	0.211	0.188	0.174	0.152	0.137	0.125	0.112
C5	0<GB<=10	6	0.32	0.295	0.26	0.237	0.218	0.202	0.189	0.178	0.169
	10<GB<=20	7	0.40	0.382	0.357	0.329	0.304	0.296	0.273	0.245	0.217
P1	0<GB<=20	4	0.2	0.193	0.179	0.167	0.143	0.128	0.117	0.109	0.093
P2	0<GB<=20	5	0.214	0.204	0.211	0.188	0.174	0.152	0.137	0.125	0.112
P3	0<GB<=20	5	0.214	0.204	0.211	0.188	0.174	0.152	0.137	0.125	0.112
P4	0<GB<=10	6	0.32	0.295	0.26	0.237	0.218	0.202	0.189	0.178	0.169
	10<GB<=20	7	0.40	0.382	0.357	0.329	0.304	0.296	0.273	0.245	0.217
P5	0<GB<=10	6	0.32	0.295	0.26	0.237	0.218	0.202	0.189	0.178	0.169
	10<GB<=20	7	0.40	0.382	0.357	0.329	0.304	0.296	0.273	0.245	0.217

maximum working voltage. The considered guard-band for the circuit increases the slack times and therefore affects the precision of the monitoring structure. The width of AI-Counters determines the maximum frequency of its input clock (i.e., RO clock) and finally, this frequency determines the precision of our monitoring method.

Table 4 reports the width of the AI-Counters and also the precisions of the monitoring system under different guard-bands and different operating voltage levels. In this study, the guard-band was defined as a percentage of the delay of the longest path. Period of the system clock of each studied benchmark circuits was equal to the delay of the critical path of that circuit plus by the defined guard-band. As results show, by increasing the operating voltage level, the precision of the monitoring system is increased (smaller RO period). However, under the considered guard-bands the width value of the AI-Counters is almost constant for P1 to P3 and C1 to C4. Results for P4 to P5 and C5 show that larger guard-bands cause lower precision, and larger counter width. Our results show that for inserting more guard-bands, we account for a little more area overhead (due to larger counters) because largest area overhead is related to the selected nodes. It should be noted that with a larger guard-band, we can decrease the number of selected nodes for monitoring, and therefore the area overhead will be reduced. This issue can be considered in our node selection and is not discussed in this paper.

In the monitoring structure of previous works, delays larger than a specific time interval (before or after rising edge of the system clock) are detected. In these works, the time interval is not adjusted based on the system aging and therefore they cannot define the aging rate. The only method that can be compared with our method is [18]. The detection precision of this work is defined according to the chain of delay elements and is almost equal to our method.

D. AGE DETECTION HARDWARE OVERHEAD

The accuracy of age delay prediction and detection methods depend on the number of paths that are selected for

monitoring. As the number of selected paths increases, prediction and detection of changes in delay will be improved. However, a large number of paths for monitoring causes an increase in the area/power overhead. The area overhead of our monitoring structure mainly depends on the number of the monitoring nodes and therefore the number of selected critical paths. This parameter is determined by the designer. As an example, Table 5 shows the area overheads of the monitoring structure for the execution unit of the Plasma processor under a different number of monitoring paths. As the results show, by increasing the number of monitoring paths, the area overhead of the monitoring structure increases. In this example, in the worst-case, the area overhead is about 3.2 percent.

The area overhead of the proposed method in comparison with state-of-the-art methods is illustrated in Table 6 under 10 percent delay guard-band and 1.2 V as the maximum operating voltage level. The reported overheads were obtained when the number of monitoring nodes was as in Table 3. Because none of the state-of-the-art works consider working under different operating voltage levels, in this comparison study, we only consider one AI-Counter in AIT (one voltage level).

In order to compare our work with other works in the literature, we use an identical path and node selection method for all works. In fact, all works are applied to the same set of critical paths. Then, we implemented the works for which we had access to their complete schematic and hardware details. For these test cases, we developed their Verilog descriptions and synthesized them using the Synopsys Design Compiler. However, for others whose structures were defined at the

TABLE 5. Area overhead (%) of the proposed structure.

#of monitoring paths(%)	10%	20%	30%	40%	50%
Part of System					
Gate overhead %	0.18	0.29	0.4	0.42	0.44
Flip-flop Overhead %	5.33	6.7	8.1	9.6	10.01
Overall overhead %	0.69	1.34	1.95	2.57	3.18

TABLE 6. Area overhead (%) of the proposed method in comparison with other methods.

Circuit Method	P1	P2	P3	P4	P5	C1	C2	C3	C4	C5	Avg.
Our method	1.88	1.09	1.70	0.86	0.59	5.12	0.72	1.02	0.29	0.12	1.339
[6]	2.04	1.2	1.873	1.04	0.65	5.05	0.91	1.6	0.4	0.23	1.499
[18]	2.25	1.307	2.04	1.03	0.71	5.48	1.07	2.48	0.61	0.302	1.727
[11]	3.06	2.51	2.89	1.873	1.24	5.6	1.49	3.01	0.82	0.347	2.284
[12]	0.24	0.68	0.72	0.298	0.312	3.05	0.41	0.89	0.08	0.05	0.673
[19]	2.43	1.301	2.012	1.053	0.865	5.7	1.36	2.36	0.65	0.373	1.810
[15]	4.37	3.4	4.1	2.2	1.83	7.45	2.5	3.44	1.43	1.08	3.18
[22]	2.54	2.05	2.173	1.84	1.27	6.41	1.36	1.78	0.67	0.4	2.049

transistor level, we estimated their hardware based on the number of gates and flip-flops. For work in [15], we implemented the selected critical paths as replica circuit. In the case of the proposed method in [11], we added the monitoring hardware at the end of each selected node. We implemented only the shadow registers of [12] and did not implement the whole controller that recovers from an error. We implemented [22], by adding the equivalent gates at the selected nodes. We did not compare our method with that proposed in [20] due to the lack of implementation details.

In comparison with the previous works, area overhead of the proposed method is acceptable. Our method provides information about the present slack time and rate of slack time changes, while the other monitoring systems can only detect that the circuit is aged or system fails due to excessive delay. The area overhead of the proposed method compared to the method with lowest (highest) area overhead is, on average, 50.26 percent (42.11 percent) larger (lower). It should be noted that the method with the lowest area overhead is Razor that only detects the age of a system after a failure. Our method has smaller area overhead than other methods. Also, based on the reported area overhead, our proposed method has, on average, 1.339 percent area overhead.

In the proposed structure, there is a constant area overhead due to the AM-Clock generator structure with an average area overhead of 0.77 percent. The remaining area overhead is related to the structure that is inserted into the selected monitoring node. For Example, for C5 0.051 percent of area overhead is related to the AM-Clock generator and 0.069 percent is related to 24 monitoring nodes.

The area overhead of the proposed monitoring system by considering different operating voltage levels are reported in Table 7. In this study, the considered guard-band was 10 percent of the critical path delay and the number of monitoring nodes was as in Table 3. Simulation results show that our proposed method has an average 2.128 percent area overhead.

It should be noted that each critical cluster has a separate AM-Clock generator that is placed near the cluster. Our experiments show that critical paths in a cluster are very close to each other and occupy a small region. For this small region, the AM-Clock does not need any buffers and acts as an extended combinational signal. However, when the potential critical paths are widespread, the AM-Clock distribution

TABLE 7 Area overhead (%) of the proposed method that considers nine different voltages.

Circuit	P1	P2	P3	P4	P5	C1	C2	C3	C4	C5	Avg.
Overhead (%)	2.74	2.35	2.62	1.46	0.98	8.14	0.9	1.57	0.33	0.191	2.128

needs buffers to retain the signal integrity, and consequently, the area overhead will increase.

E. MONITORING STRUCTURE POWER CONSUMPTION

Another important issue in the monitoring structure is its power consumption. We estimate the power consumption using switching activity (extracted from gate-level simulation) and load capacitor (extracted from HSPICE) of each node of the benchmark circuit and the monitoring hardware. However, we obtain the power consumption of the proposed monitoring structure and also the benchmarks using PrimeTime. Results show almost the same ratio of power overhead.

The power overhead of the proposed method is illustrated in Table 8 under 10 percent delay guard-band and 1.2 V as the maximum operating voltage level. The reported overheads were obtained when the number of monitoring nodes was as in Table 3. The power consumption of the monitoring systems was determined when all of their components are operational. This study shows that the monitoring structure increases the power of the benchmark circuits by only about an average of 0.69 percent.

Our experiments show that our method consumes power smaller or equal to other works, and like other works in this area, its power overhead is low. The overhead of our method is about 1.01 percent more than the lowest one ([12]), and about 3.2 percent less than the highest one ([14]). This study shows that the power overhead of the age monitor structure is not much compared to the whole power consumption of the circuit. Also, the monitoring mechanism can be activated only at specific times to reduce the overall power consumption. Since we calculate the average power consumption of the monitoring circuit with the system that we are monitoring, the energy overhead of the monitoring circuit has the same respective ratio as the overall power overhead. We estimate the power of the system that is enhanced with monitoring hardware and compare it with the system without monitoring hardware. We consider the same working duration for both systems. Therefore, the energy overhead is same as the reported power overhead.

F. EFFECTS ON SYSTEM PERFORMANCE

In the proposed structure, a flip-flop is added to the output of the selected monitoring nodes which may impact on the

TABLE 8. Power overhead (%) of the proposed structure.

Circuit	P1	P2	P3	P4	P5	C1	C2	C3	C4	C5	Avg.
Overhead (%)	0.8	0.704	1.217	1.08	0.35	1.60	0.29	0.632	0.14	0.091	0.69

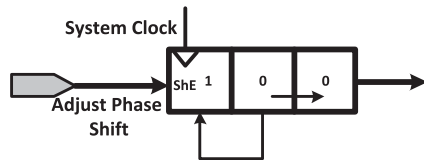


FIGURE 15. Structure of the noise prevention unit.

delay of the path. Our studies show that in the considered benchmark circuits, this load leads to 0.43 to 1 percent delay increase, which is almost negligible. Prior age monitoring methods also add sensors or flip-flops at the selected monitoring nodes which increase the load capacitance of the selected nodes. Hence, this effect is a normal consequence of monitoring and cannot be avoided. The performance loss only occurs for longest paths that one of their nodes is selected for monitoring. However, our node selection method tries to find the nodes that are not in the longest paths, and covers them as mentioned in Section IV.

VI. CONSIDERING NOISE, SUPPLY VOLTAGE, AGING, AND TEMPERATURE

In this section, some phenomena that can affect the proposed monitoring structure are discussed.

A. NOISE AND SOFT ERRORS

Noise is a common problem of age monitoring methods such as Razor [12], [13], because all of them use flip-flops for capturing arrival value before or after the active edge of the clock. Noises such as soft-error can affect the accuracy of these techniques. Our work can also suffer from this issue. It may reduce the accuracy of our method. When a noise causes issuing a false APS signal, it will be interpreted as an early aging signal. This can cause problems in the unlikely situation of multiple consecutive issuances of the APS signal. Eventually, however, the age of the system, will catch up with the early erroneous aging signals. To avoid false aging reports in noisy systems, we can add an optional noise prevention unit that would only trigger the age monitor controller if the APS signal is issued twice in a row. This leads to reducing false alarms of the APS signal due to soft noise. The structure of this unit is shown in Figure 15. It consists of a ring-counter that is initialized with “100”. If the APS signal is issued, this ring-counter shifts to right and will be “010”. The second pulse on the APS signal causes storing “101” in the ring-counter. The ‘1’ in the least significant bit causes halting the AM-Clock generator and adjusting it according to the system’s age.

B. POWER SUPPLY VARIATION AND VOLTAGE DROP

Simultaneous switching or voltage drop is also a common problem of age monitoring methods. As indicated in [28], “voltage drops are typically less than 10 percent of the power supply voltage”. Simultaneous switching (or voltage drop) is not a permanent issue and it occurs in the specific situations (specified activities and input patterns). We discuss the effect of voltage drop for three scenarios as follows.

Scenario 1: If voltage drop affects both system and age monitoring circuit simultaneously, it cannot disturb the monitoring mechanisms. For example, suppose in working voltage of 1.1 V, a voltage drop occurs and the voltage changes to 1.0 V. In this case, the counter that is related to 1.1 V is used that has a larger value than that of the 1.0 V. Therefore, the voltage drop cannot affect the recorded age of the system. In this case, there will be no false alarm. However, increasing of the delay of the AM-Clock generator reduces the precision of the monitoring system and causes coarser measurement intervals.

Scenario 2: If voltage drop occurs only in the circuit, the delay of selected paths is increased and if it exceeds the age measurement precision, the APS signal is issued and this increase in delay is considered as system’s age. Our proposed method deals with this situation like a noise-based error, which will resolve itself in the next aging reports. In addition, the structure of the noise prevention unit (Figure 15) is an optional way that helps to handle this case.

Scenario 3: The case that the voltage drop occurs only on the age monitoring circuit is the last scenario. This can increase the delay of AM-Clock generator, which reduces the precision of the monitoring system, causing coarser measurement intervals. Such measurements create latency in age recording. However, since the voltage drop is not permanent, the AM-Clock generator returns back to its normal frequency, and correct age reporting resumes. In our work, changes in the measurement precision for all working voltages is calculated, and the experiments show that the precision changes is about 0.018(ns) for 5 percent voltage drop and about 0.033(ns) for 10 percent voltage drop, on average.

It should be noted that the chance of scenarios 2 and 3 (above) to actually happen is relatively small. This is because the AM-Clock generator uses a very small amount of hardware and it is placed near the critical cluster, and voltage variations affecting the circuit also affect our monitoring hardware.

C. AGING

One of the issues that may affect the functionality of a monitoring structure is aging of its components. This is a problem that all age monitoring methods suffer from. In our proposed method, the only parts of the monitor structure that are active at all times are Phase-Shifter and RO. The continuous operations of these parts may cause them to age due to EM and HCI. While, other parts may suffer from NBTI phenomenon. Thus, some transistor level mechanisms will be helpful to prevent aging of these components. Mechanisms such as one proposed in [29] can be used for age monitoring hardware. However, because of the area overhead issues, these solutions cannot be employed to the whole circuit. In addition, periodical activation of the monitor could reduce its aging. In our proposed method, the age monitoring can be activated at specific time points according to the temperature of the system or the workload in order to slow down its aging. We

measure the age of monitoring structure using MOSRA in HSPICE. First, we consider nonstop operation and calculate its aging. Second, we consider 60 minutes working, 60 minutes power off and calculate the age of the AM-Clock generator. Experiment results specify that the aging advancement of the monitoring hardware after 10 years is about 1.7 percent (very low) when it operates periodically, and is about 7.4 percent in the nonstop operation.

D. PROCESS VARIATION

Another issue that can affect the aging measurement of the monitoring methods is process variation. Our experiments show that the effect of process variation on the precision of AM-Clock is about ± 8.5 percent when a nine steps RO is utilized. This changes the period of RO by about 0.034 ns, on the average. If the process variation increases the delay of the monitoring hardware, monitoring precision will be reduced. In this case, the amount of shift phase (rising edge of AM-Clock) is more than the at-birth delay of critical paths and the precision of aging steps is more than the condition without process variation. So, this case does not disturb the aging measurement.

On the other hand, if the process variation decreases the delay of monitoring hardware (i.e., higher frequency of RO), the precision of the proposed method will be improved. In this case, the amount of shift phase (the rising edge of AM-Clock) is less than the at-birth delay of critical paths and this results in false issuing of the APS signal at first hours of system's operation. However, the monitoring structure will be adjusted accordingly. The precision of aging steps in this case is more than the condition without process variation.

In order to immune RO from the process variation, one can optionally use the structure of [30] for the process variation independent RO in the proposed AM-Clock generator. This increases the overall hardware overhead.

E. HOTSPOT

Hotspot is an inevitable issue in digital systems. The delay of paths can change due to the high temperature of hotspot. After a hotspot, when the temperature decreases, the delay of the critical paths return to their normal states (although, in the long term, the temperature increases the age of the system that can be detected by the monitoring structure). Our experimental results show that change in temperature from 27 °C to 127 °C does not change age counter values (part of these results are shown in Figure 10(c)). Therefore, if hotspot affects both the monitoring hardware and the critical cluster of the system, it cannot disturb the monitoring measurement.

If the hotspot only affects the circuit being monitored (the probability of this condition is very low, because the AM-Clock generator hardware is very small and is placed near the critical cluster), the monitor considers the changes in delay as system's age and therefore, updates the counter value. When the hotspot is resolved, the path delay values go back to their initial states before hotspot, keeping the monitoring counter value in the new overestimated state for a

relatively long time. The actual age of the system eventually catches up with this overestimated value. In effect, our proposed method deals with this situation like noise-based error.

In the case that the hotspot only affects the AM-Clock generator structure, it cannot disturb aging measurement. This only slightly reduces the precision of our measurements. The probability of this condition is also very low because the AM-Clock generator is very small and is placed near the critical cluster.

F. POWER-SUPPLY DISTRIBUTION

Vdd changes due to power supply distribution may cause voltage drop and therefore increases path delays. If the voltage drop due to power supply distribution is perdurable, then the monitoring structure adjusts itself based on the increased path delay. Because the AM-Clock generator hardware is small and can be placed near the selected critical paths, the voltage drop due to power supply distribution is the same for the circuit being monitored and the monitoring structure. Therefore, the monitoring hardware is adjusted based on the delay of the paths. It should be noted that this will be handled with the counter value of the higher voltages. Considering a 10 percent voltage drop due to power supply distribution leads to about 0.033(ns) change in the precision of detection.

VII. CONCLUSIONS

In this paper, we have proposed an age detection and prediction method that uses adjustable monitors that utilize negative-shift phase clock. The proposed method adapts itself to process and environmental variations. The mechanism of monitors, placement of the sensors in the circuit, frequency-voltage adaptability, and providing aging rate are some of the unique features of this work. Our proposed mechanism is an online method that predicts aging before it actually occurs. Our system handles multi-voltage domains by means of dedicated counters for allowed voltages. This monitoring system is self-adjusting, in the sense that dynamic and temporary environmental and circuit variations will be compensated for after the temporary situations have disappeared.

Our experimental results show the age detection precision is between 0.12 ns to 0.401 ns time interval for different voltages and for studied benchmarks. Area and power overhead of our method are negligible and are about 2.13 and 0.69 percent respectively. The performance overhead of the proposed method is low.

Other applications of the proposed architecture in this paper are in the areas of reliability, system performance, and testing. The additional hardware we insert in the system for age detection, can easily be used for offline small-delay defect testing.

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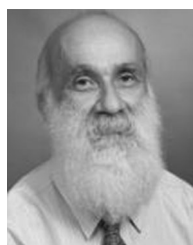
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